

Week 8 Report

16 - 20 August 2026 | UFACTORY xArm 6 work cell

The Arm

I started a fresh sorting cell - camera on a side stand, suction cup, blue cubes picked and red ones left - and found why the previous version could never pick. The link between what the camera sees and where the arm must go was off by about 17 mm, against roughly 6 mm of tolerance. The cause was the calibration being done by hand: the vacuum sensor confirms the cup sealed but cannot tell whether it was centred. The replacement lets the arm hold the cube itself and drive through a grid of positions, which removes the hand from the measurement entirely. It is built and tested offline, but not yet run on the arm.

Documentation

I wrote a full report of all six projects built on this arm, finished the xArm 6 guide book in the KACST format, and produced the King Saud University co-op presentation from the training report.

Next

Run the new calibration on the arm and check it against a tape measure, then teach the table surface and attempt the first pick.